



Society of Petroleum Engineers

SPE-229506-MS

GenAI Multi-Agent Retrieval Augmented Generation for Oil & Gas Applications

T. Hnot, H. Nasr, and D. Zikrach, SoftServe Inc

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229506-MS](https://doi.org/10.2118/229506-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Large Language Models (LLMs) have emerged in recent years as one of the most significant advancements in artificial intelligence. These models enabled the development of intelligent chatbots capable of answering a wide range of user queries by referencing vast amounts of information. However, LLMs have many limitations. Their responses can be unpredictable, often lack domain-specific knowledge, and may include contextual misunderstandings and hallucinations. To address these challenges, Retrieval-Augmented Generation (RAG) has become a prominent technique. RAG systems decouple world knowledge from the model's parameters, combining the generative capabilities of LLMs with advanced information retrieval, processing, and vector storage techniques. These integrations result in responses that are more accurate, contextually relevant, and significantly less prone to hallucination.

More recently, Multimodal Large Language Models (MLLMs) have extended LLM capabilities beyond text to include modalities such as images, tables, videos, and charts. These multimodal capabilities are critical for domains like oil and gas (O&G), where data exists in diverse formats. Traditional RAG systems often rely on static pipelines, which limit their effectiveness in handling multimodal queries. However, agentic-based RAG architectures offer a promising solution by enabling dynamic and adaptive processing.

In this paper, we introduce a novel GenAI-based multimodal agentic AI assistant for drilling and completion applications (D&C). This domain known for data complexity and variability. By analyzing historical wells' D&C reports, wells' performance, geological characteristics, and production trends from nearby wells, the system enables engineers to identify optimal drilling plans and improve field development.

It demonstrates that the GenAI assistant performs well in handling complex, domain-specific questions that required synthesizing several types of data. A series of experiments show significant improvements (contextual understanding, hallucination prevention, and overall reliability) compared to traditional approaches.

Introduction

O&G companies gather massive amounts of data across their value chain, some of this data must be stored for many years. O&G data is structured and unstructured, and multimodal in nature, it contains text, time-series data, videos, images and charts. O&G companies use historical data to support intelligent

decision making in planning and daily operations. Information about past assets represents a critical part of any company's knowledge base, especially as the industry struggles to retain knowledge from an aging workforce, whose leading experts are nearing retirement.

Knowledge management has been a key focus for the industry since the 1990s. Many initiatives in this area have had limited success. Generative AI, Large Language Models (LLMs) and Retrieval Augmented Generation (RAG) technology can help the industry accelerate knowledge management, enabling companies to make effective use of their massive historical data and to capture and retain the knowledge of domain experts.

There are many applications in O&G that can leverage these technologies. Drilling and completion (D&C) is one of such important applications. As a typical asset, companies have hundreds or thousands of reports from multiple wells gathered over the years. These reports are critical for companies to make future field-development decisions.

System Solution Overview

In this paper we present a novel GenAI assistant that addresses the multimodality of D&C application and incorporates an agentic AI approach. In the forthcoming sections, we present detailed system architecture and results from comprehensive testing and benchmarking. The solution comprises three key components:

Multimodal LLM. The model can perform a wide range of cognitive tasks:

1. Processes multimodal data and delivers advanced analyses of complex inputs.
2. Can extract information from visuals such as charts and diagrams, and automate repetitive tasks and operations.
3. Can process multimodal data and delivers advanced analysis around complex data structures such as tables.
4. Has the capability to extract data from visuals like charts and diagrams, and automate repetitive tasks and operations.

Vector Database. A vector database allows users to store and search vector embeddings generated by LLMs. These embeddings capture the semantic meaning of complex data such as text, tables, images, video or audio. By enabling fast and accurate similarity searches, vector databases are essential for powering use cases like information retrieval, insights generation, summarization, personalized recommendations, etc. Their ability to efficiently retrieve relevant information based on context makes them a key component in modern AI applications.

Agents. Multiple specialized agents can collaborate on various tasks and process different types of data to provide a comprehensive answer to queries. The agent module can use foundational models and various APIs to connect different sub-systems. Agents decompose user requests, collect relevant information, and efficiently complete tasks, while performing memory retention as well as security and reliability checks. Agents break down query processing into specialized components like semantic parsing, cross-modal retrieval, and context-aware generation, allowing targeted optimization while maintaining overall adaptability.

Tests and Benchmarking

Several tests were conducted to evaluate the developed system. The test results were benchmarked using multiple evaluation metrics, such as:

1. Answer Correctness - reference-guided pairwise comparison to measure the correctness of the generated answer (LLM-as-Evaluator).
2. Answer Relevancy - assessment whether the generated answer addresses the user's query.

3. Faithfulness - consistency between the generated answer and the retrieved textual context (to check whether LLM is not making up information).
4. Contextual Relevancy - quality of the retrieved context in supporting the answer.

Benefits to Drilling and Completion

Drilling and completion data represents some of the most complex scenarios for LLMs and RAG systems. As illustrated in Figure 1, D&C data is inherently multimodal: text, tables, charts, seismic images (2D and 3D), well logs visuals, maps (geospatial and reservoir), production data, educational videos. GenAI assistants allow engineers efficiently analyze historical D&C data from nearby wells, evaluate subsurface conditions, and design drilling plans that maximize recovery.

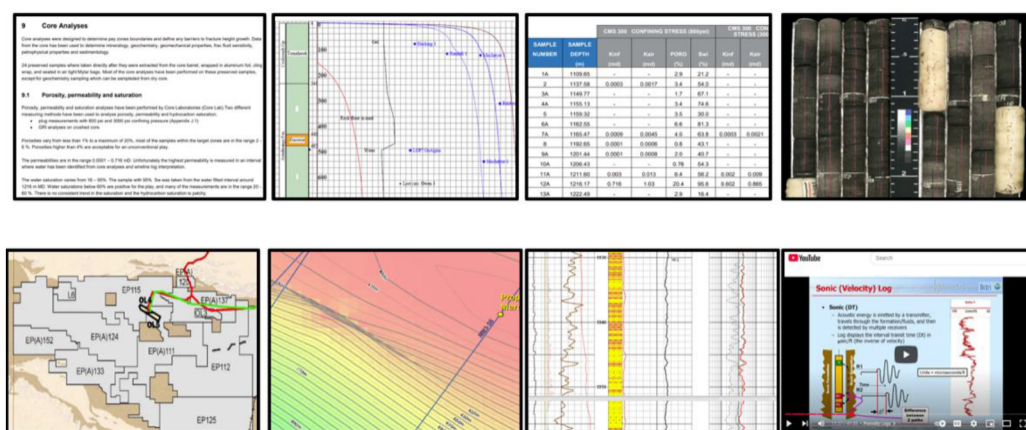


Figure 1—Example of multimodality of drilling and completion data

Field-development engineers need to closely examine historical wells' performance, geological characteristics, and production trends, to identify the most promising drilling locations, improve well placement in production zones and maximize recovery. Nearby well data provides crucial geological context for understanding the subsurface in the area, potential risks and opportunities of a newly proposed well. Data from nearby wells provide insights into reservoir characteristics such as fluid characteristics, porosity, permeability, and other factors that support in reservoir characterization, maximize reservoir contact and recovery.

Nearby well information helps identify potential challenges and refine drilling strategies by predicting issues such as wellbore instability, lost circulation, or stuck pipe incidents, characteristics of the reservoir, including properties like porosity and permeability.

GenAI-based assistants can be essential tools to help engineers in:

1. **Well Drilling Planning** - by analyzing offset well data, GenAI tools help refine well designs and trajectories. Engineers can draw on the best practices and anticipate challenges based on previous drilling experiences in similar geological formations. This can help minimize non-productive zones and maximize reservoir contact.
2. **Reducing Risk and Improving Geo-steering** - historical drilling data provides important insights on potential geological risks related to faults and fractures, engineers can incorporate this information into pre-drill planning and real-time geo-steering to help mitigate uncertainties and reduces the likelihood of encountering unexpected challenges during drilling. Using correlation with adjacent wells during drilling and real-time data from Logging While Drilling (LWD) tools, drilling operators can adjust the drilling path to optimize the well trajectory.

3. **Enhancing Reservoir Characterization** - nearby wells seismic surveys, LWD data, mud logging data, core samples and other wellbore measurements constitute a rich dataset for reservoir characterization. GenAI assistants can aggregate and analyze these multimodal inputs to estimate key reservoir properties such as porosity, permeability, and fluid saturation. This provides contextual knowledge, which is essential for optimizing well placement and maximizing production potential.

Related Work

The release of ChatGPT-3 in 2020 ushered in a new era in Artificial Intelligence. Large Language Models are trained on massive amounts of data, enabling them to generate human-like responses during interactions. This capability has positioned generative AI systems at the forefront of innovation across various domains, where they are increasingly used to build intelligent assistants and decision-support tools to tackle complex problems.

However, LLM technology has many limitations. LLMs are trained on static datasets and often lack accuracy in many applications; systems tend to exhibit hallucinations, lack contextual understanding, specificity and domain-specific expertise.

RAG emerged as a promising technology and solution to address the limitations of static knowledge bases in traditional generative models by leveraging advancements in LLMs and vector retrieval technologies, as shown in Figure 2. RAG operates on a simple principle: augmenting the generation process with relevant information from external data stores. This approach enables the system to access and utilize a vast repository of knowledge that can be easily updated and expanded.

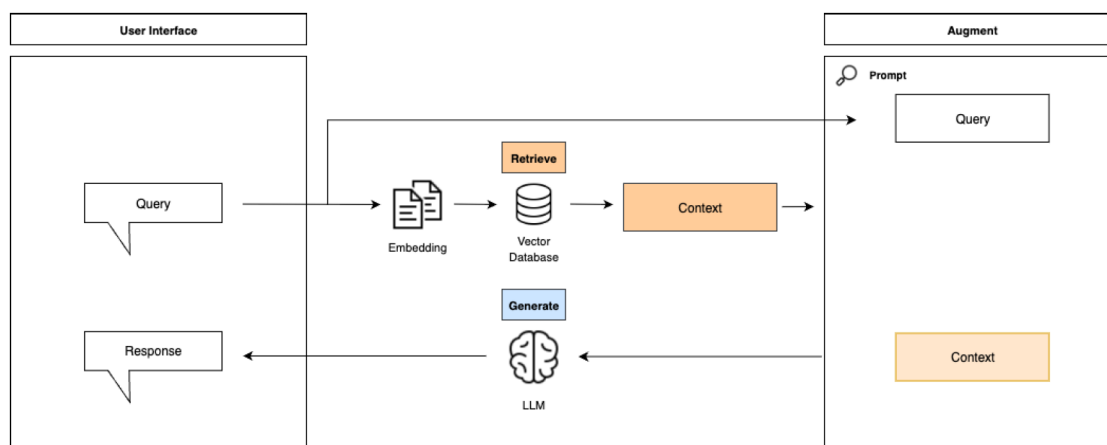


Figure 2—Basic RAG architecture

Meta AI introduced the first RAG framework in 2020, Figure 3, Lewis et al. (2021) in a paper titled "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" to augment generative models with external knowledge sources to improve the performance and address limitations of LLMs. Guu et al. (2020), also published early research work on RAG systems. The basic concept proposed to decouple world knowledge from the model's parameters. More specifically, these models are trained to access an external memory to enhance the model's predictions. A comprehensive survey by Gal et al. (2023) traces the evolution of RAG techniques and highlights their effectiveness on complex reasoning tasks, consistently outperforming traditional LLM chatbots.

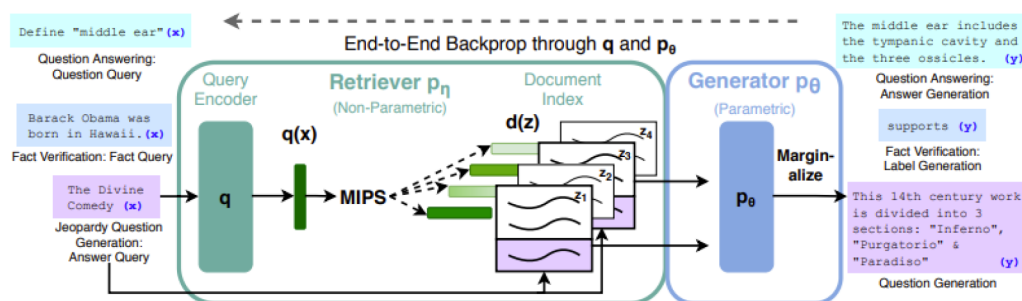


Figure 3—First RAG architecture proposed by Meta

The key benefits of RAG systems include:

- **Improving Response Accuracy:** By referencing external knowledge bases, RAG can provide more accurate and up-to-date responses compared to models relying solely on their static training data.
- **Incorporating Contextual Relevance:** RAG's retrieval mechanism ensures that responses are tailored to the user's specific query and context, making them more informative and engaging.
- **Reducing Hallucinations:** RAG helps mitigate the risk of hallucination by grounding responses in factual data retrieved from reliable sources.

Despite rapid progress, RAG systems still face problems. Answers quality is bounded by the recall and precision of the retriever, and unreliable generation components can introduce factual drift or formatting errors [Wu et al. \(2024\)](#). Many early RAG implementations also assumed purely textual data, limiting their applicability to domains that heavily rely on multimodal content.

Multimodal RAG

Many industrial scenarios, including Oil&Gas, manufacturing, finance, and healthcare, store knowledge in multimodal formats: text, tables, charts, images, video, audio, geospatial maps, and complex unstructured data. The introduction of Vision Language Models (VLMs) and Multimodal LLMs (MLLMs) by OpenAI and others was key in helping to advance multimodal RAG systems to enable more comprehensive and accurate responses by integrating multiple data types, [Yin et al. \(2023\)](#), [Zhang et al. \(2024\)](#), [Sun, L., et al. \(2024\)](#). As a result, GenAI systems can now respond to complex queries and can retrieve an image alongside text to help answer a question that requires visual context. The generative model then synthesizes both the retrieved textual and non-textual information to generate a comprehensive and multimodal response. This advancement allows Multimodal RAG systems to handle more complex tasks that require understanding and integrating different forms of data. Applications and research in Oil&Gas applications are very limited; the literature search resulted in work only related to industrial applications [Aquino, G. et al. \(2025\)](#).

To overcome these challenges, [Drushchak et al. \(2025\)](#) describe the mRAG architecture that can ingest and reason over text, tables, images, and video. The authors report that this unified design increases answer accuracy and dramatically lowers hallucination rates compared with text-only RAG, underscoring the value of multimodal grounding for knowledge-intensive domains.

Agents in Multimodal RAG. Traditional RAG implementations often rely on static pipelines that struggle with multimodal query processing. Recent agent-based RAG architectures, [Figure 4](#), [Cheng, Y., et al. \(2024\)](#), [Aquino, G., et al. \(2025\)](#) have addressed these limitations by improving system modularity and operational flexibility. The agent-oriented approach breaks down query processing into specialized components like semantic parsing, cross-modal retrieval, and context-aware generation, allowing targeted optimization while maintaining overall adaptability.

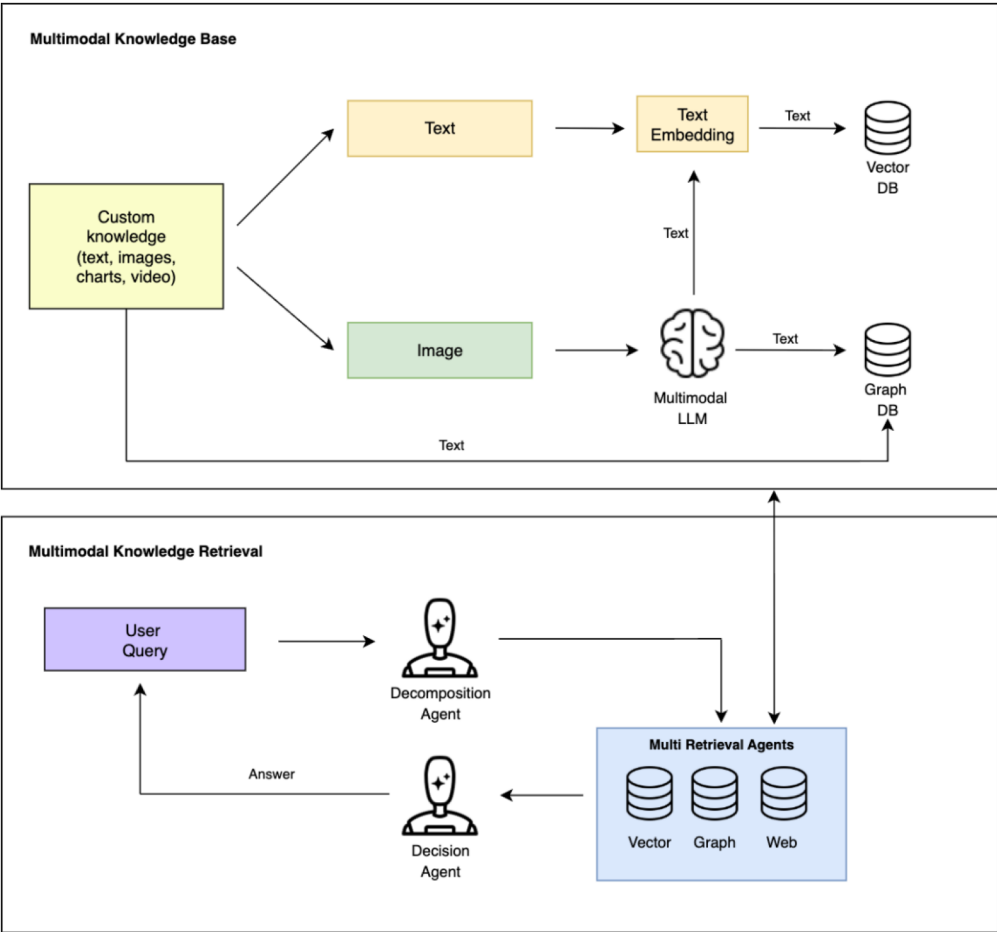


Figure 4—Basic concept of multimodal agentic RAG architecture

Methodology

To contextualize the proposed system architecture, we contrast the capabilities of traditional RAG pipelines with the more advanced multimodal RAG (mRAG) approach adopted in this work.

Table 1—Comparison of Traditional RAG and Multimodal RAG

Aspect	Traditional RAG	Multimodal RAG
Data Type	Primarily processes textual data.	Handles multiple modalities, including unstructured (text, images, schematics, tables, videos) and structured (databases, tables).
Accuracy and Insights	Suitable for basic use cases involving textual content. Struggles with complex data such as images or dense tables.	Offers richer, more context-aware insights by combining multiple data types and understanding the semantic interplay between them.
Speed of Response	Fast response times due to straightforward text retrieval.	Slower initial response due to preprocessing but benefits from deeper understanding and richer indexing.
Intelligent Processing	Works well for simple inputs but often requires manual handling as complexity increases.	Scales with complexity; integrates structured and unstructured content and supports intelligent reasoning through embeddings.
Summary	Effective for unimodal use cases involving simple documents and FAQs.	Ideal for scenarios requiring deep understanding across diverse data formats, such as technical manuals, diagrams, or transcripts.

This comparison highlights the fundamental motivation behind building the proposed system architecture: the multimodal RAG approach allows for more robust and intelligent information retrieval and generation, which is critical for tasks involving rich technical documentation and visual data.

Dataset Description

The solution was deployed and tested in production on dozens of pdf documents. To run the evaluation, we shrunk the data and used a dataset of:

1. 12 open-sourced completion and drilling reports (regarding wells: Carpentaria-2 / 2H, Tanumbirini 3H / Tanumbirini 3HST1, OzAlpha-1., etc.) with 874 pages in total (including 547 images and 1025 tables).
2. 4 videos that describe well-logging process, its interpretation and petrophysics introduction with total duration of 83 minutes.

System Architecture

The system architecture was developed based on the design proposed by [Drushchak et al. \(2025\)](#), who introduced an mRAG framework capable of ingesting and reasoning over text, tables, images, and video.

The overall system architecture is organized into three primary layers, as illustrated in [Figure 5](#):

- **Data Processing:** responsible for initial input preparation and cleaning;
- **Embedding and Indexing:** transforms multimodal data into semantic representations and performs efficient indexing;
- **Retrieval and Generation Engine:** handles querying the indexed data and generating contextually relevant responses;
- **Hybrid Search:** improves retrieval quality across diverse data modalities and content structures.

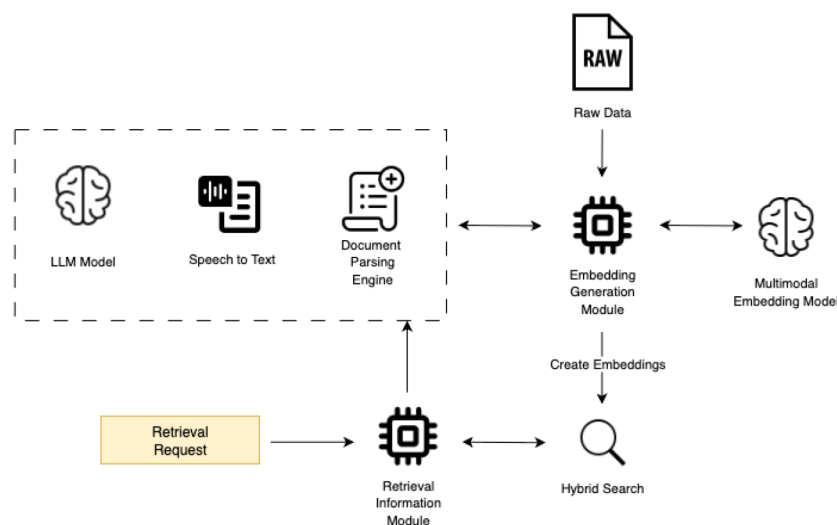


Figure 5—The overall architecture of the solution

Hybrid Search. The hybrid approach was particularly beneficial when processing text and table data, where domain-specific terminology plays a key role, and in video transcripts. Hybrid search combines **vector-based retrieval** (semantic search) and **sparse keyword-based retrieval** (lexical search), enabling the system to leverage the strengths of each method. Semantic search captures the underlying meaning of queries and content, allowing for effective retrieval even when there is little or no lexical overlap. This is particularly useful in multimodal contexts, where image captions or transcribed speech may not contain

exact keyword matches. On the other hand, lexical search focuses on exact or near-exact matches between the user query and document content. This method excels in scenarios where precise terminology, named entities, or keywords carry critical meaning. In the context of our system, lexical search played a crucial role in maintaining precision, ensuring that important query terms were not overlooked by a purely semantic model. By integrating these two methods, the system can achieve a better balance between generalization and specificity, resulting in more accurate and relevant retrieval outcomes. The overall hybrid search process is illustrated in [Figure 6](#).

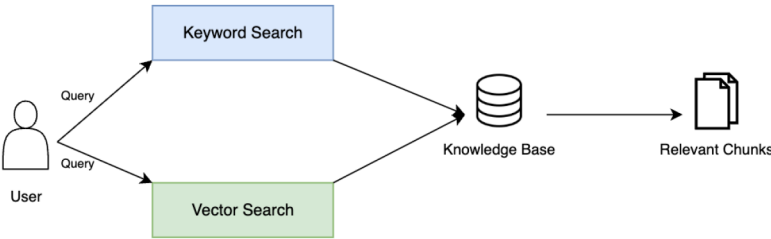


Figure 6—Hybrid Search

Information Retriever

This module prepares data from both document and video sources for semantic retrieval. Below, we outline the pipelines used for each modality.

PDF-Based Retriever. This pipeline extracts and semantically indexes text, tables, and images from PDFs for retrieval purposes. The pipeline is shown in [Figure 7](#).

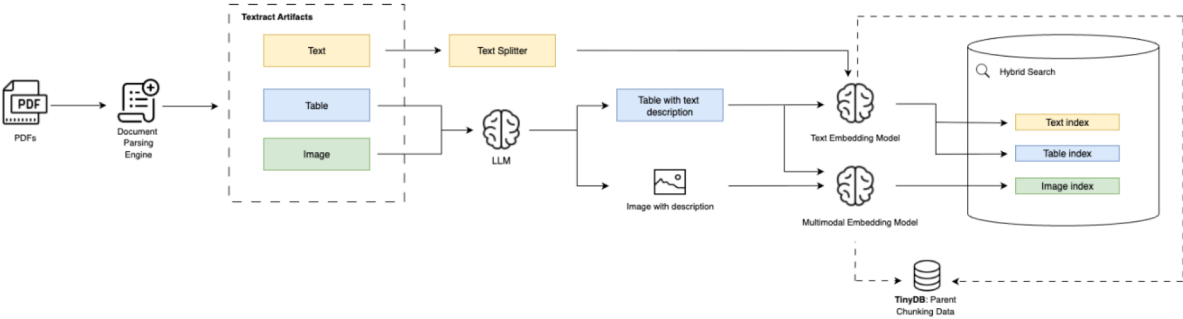


Figure 7—PDF-Based Retriever architecture

Overview of the process:

- 1. Document Parsing Engine: Extracts text, tables, and images from PDFs;
- 2. Text Splitting: The text is divided into meaningful segments based on contextual relevance;
- 3. Table Processing: LLM produces concise semantic representations of tabular information;
- 4. Image Processing: LLM creates contextual descriptions for each extracted image;
- 5. Embedding and Indexing: Descriptions are encoded via a text embedding model and multimodal embedding model and then indexed into a vector search engine for downstream retrieval;
- 6. Metadata Tracking: A database - TinyDB (2025) stores tables, images, and indexing metadata such as parent-child relationships between data chunks.

Video-Based retriever. The video pipeline processes multimedia files to generate searchable representations of keyframes and transcript content. The pipeline process is illustrated in [Figure 8](#).

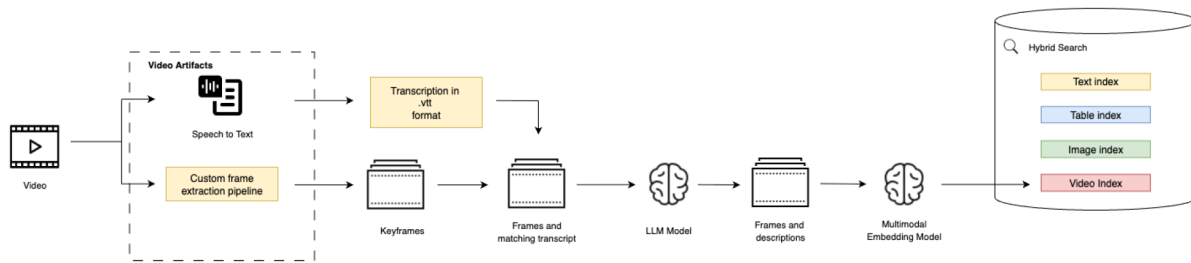


Figure 8—Video-Based Retriever architecture

Overview of the process:

1. **Speech Transcription:** A speech-to-text engine transcribes video audio into time-aligned captions.
2. **Keyframe Extraction:** A custom pipeline based on OpenCV (2025) extracts keyframes by detecting scene changes and analyzing content. The parameters applied for entropy, edge ratio, contrast, and pixel changes follow those established by Drushchak et al. (2025).
3. **Frame-Transcript Alignment:** Keyframes are paired with the corresponding transcript segments based on time-window alignment.
4. **Visual Description Generation:** LLM produces detailed descriptions of keyframes by combining visual elements with the relevant transcript context.
5. **Multimodal Indexing:** The enhanced keyframe representations are encoded using a multimodal embedding model and organized within a vector database under indexes specific to each video.

Multimodal Retrieval Augmented Generation

This section presents the fundamental components of the mRAG system designed to respond to user queries based on file data.

1. **User Query Processing:** LLM analyzes the user's input and, depending on the context, either reformulates the query to maintain conversational continuity or retains the original query if it is unrelated to previous exchanges.
2. **Independent Retrieval:** Relevant chunks of text, tables, images, and videos are retrieved through embedding similarity searches. Smaller embedding-based chunks are used for searching, while larger parent chunks are supplied to the model for context. For video data, retrieval relies solely on embeddings. The system selects the top 10 textual chunks and the top 5 chunks from other modalities.
3. **Answer Generation:** The selected data, together with the original query, is fed into the LLM, which produces a coherent and context-aware response.
4. **Citation and Traceability:** To maintain transparency and enable verification, all sources are properly cited with links pointing to specific document pages or video timestamps

Monitoring and Feedback Loop

Each question was evaluated by executing the evaluation script three times to ensure consistency. To enable observability, reproducibility, and data-driven insights essential for diagnosing model performance, Langfuse (2025) and DeepEval (2025) were utilized. Langfuse was used to monitor and trace all interactions with the model. It generated detailed logs of inputs, outputs, latencies, and metadata, allowing evaluation sessions to be tracked over time, specific cases to be inspected, and different model behaviors to be compared. DeepEval served as the structured evaluation framework, enabling the definition of custom metrics such as relevance, factual correctness, and coherence.

Insights Engine Application

The GenAI Insights Engine was developed to provide end-users with an interface that allows them to send queries and receive multimodal outputs (the system delivers responses containing images, graphs, tables, and text). By design, each chatbot response consists of three main components:

1. Textual response to the user's query.
2. Related materials section, which includes tables, images, and other visuals that enhance the answer.
3. Citations section, listing the sources used to generate the response.

Below are a few examples of how different queries are handled by the GenAI Insights Engine solution:

1. If only **Text** chunks are among the closest matches to the query based on embeddings, the system will generate a response using these chunks and include them as citations. This represents the typical scenario of a classical RAG (Retrieval-Augmented Generation) chatbot system (Figure 9).
2. **Image** modality will appear in the response when description of image generated by multimodal model combined with the nearest text would be close to user query (Figure 10). Such responses provide visual context, which is often helpful for gaining a more comprehensive understanding of the requested content.
3. **Table** modality will appear in the response when description of table generated by multimodal model combined with the content of the table itself would be close to user query (Figure 11). This modality is one of the most important taking into account nature of much of data and the way how structured representation may improve information perception.
4. The **Video** modality is not used as frequently but can enhance responses by providing direct references to relevant video recordings along with corresponding timestamps. The search mechanism is similar to that used for images, relying on a combination of key video segment descriptions and relevant transcribed text to determine relevance to the user query (Figure 11).

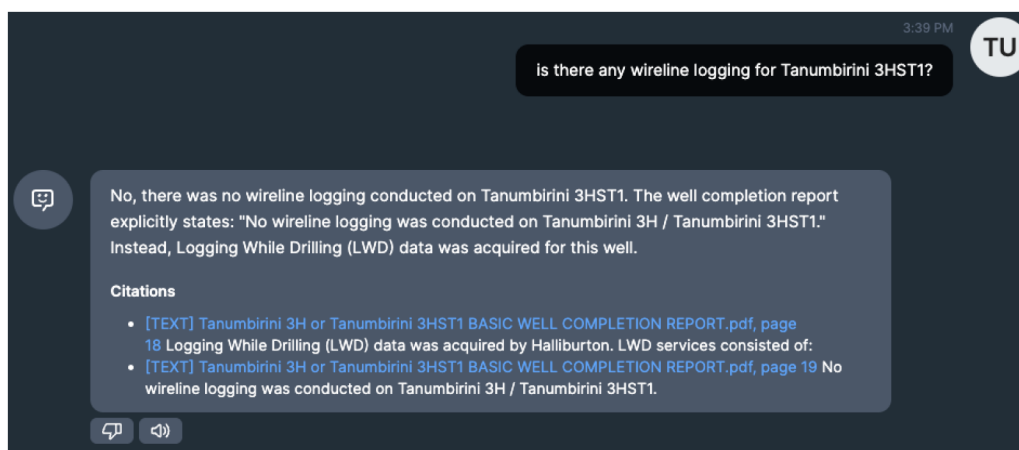


Figure 9—Text-based Retrieval in GenAI Insights Engine

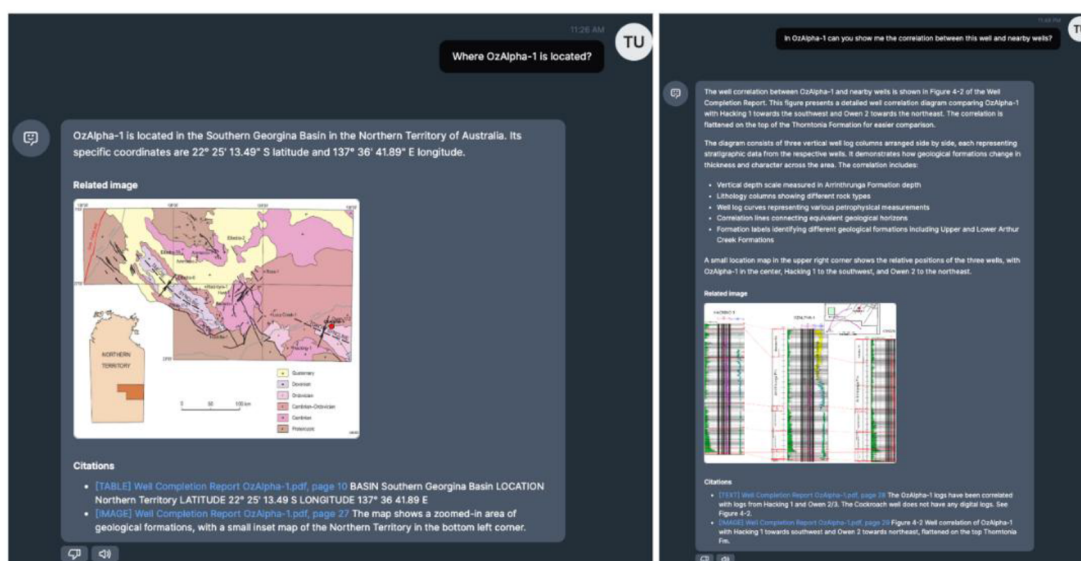


Figure 10—Text-based Retrieval in GenAI Insights Engine

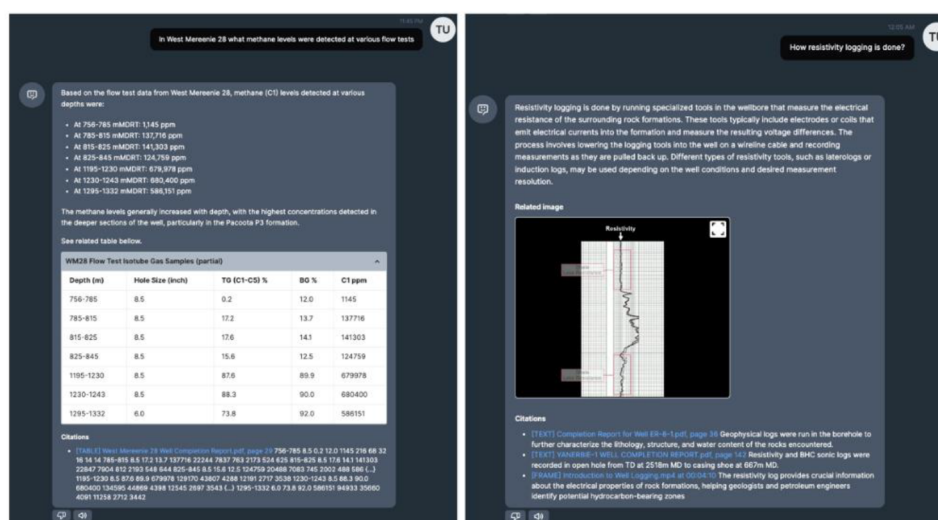


Figure 11—Table-based and Video-based Retrieval in GenAI Insights Engine

Experiments

Experiments Intro

We conducted two experiments:

- The first experiment involved comparing traditional RAG (Azure OpenAI Assistant) and mRAG. With the evolution of LLM, there is a growing need to create systems that allow the user to receive answers based on their documents. This has led to the RAG (Retrieval Augmented Generation). RAG combines three components - retrieval (query is converted to embeddings and then the most relevant documents for this query are retrieved), augmentation (the final prompt with query and documents is created) and generation (LLM processes the prompt and generates a response);
- The second experiment involved benchmarking the multimodal RAG pipeline using five language models.

Experiments Setup

For evaluation, we used a dataset consisting of 12 PDF documents and 4 videos related to energy specifics. The benchmarking set includes 43 questions: 13 for text, 11 for tables, 2 for images, 2 for videos, and 15 for mixed types (where the answer requires information from multiple modalities). Each question was evaluated by running the evaluation script three times. For experiments we used Langfuse and DeepEval (2025) as core evaluation framework. An example question format is:

```
{
  "query": "Provide the summary of operations for OzAlpha-1.",
  "answer": ""
  OzAlpha-1 was an exploration well drilled by Statoil in the South Georgina Basin, Australia. Key
  operations included:
  Spudded on April 1, 2014 and completed in 20 days.
  Drilled to a total depth of 1250.3 m MD.
  Used a two casing string design:
  12 1/4" hole drilled to 507.5 m, 9 5/8" surface casing run and cemented.
  7 7/8" hole drilled for lower section.
  Continuous coring through target Arthur Creek Hot Shale and Thornton Limestone.
  Wireline logging performed.
  4 1/2" production casing run to 1240 m MD KB and cemented to surface.
  Well temporarily suspended on April 21, 2014 awaiting stimulation and testing.
  ""
  "sources": ["Well Completion Report OzAlpha-1.pdf"],
  "type": "Table"
}
```

Evaluation Metrics

We evaluated model performance using both retrieval-level and response-level metrics. Table 2 presents the list of metrics and their descriptions.

Table 2—Experimental results across different modalities comparing different models.

Category	Metric	Definition
Retrieval	Contextual Precision	Check how well the answer is based on the context provided and does not go beyond it
	Contextual Recall	Check whether the answer covers the maximum amount of useful information from the context
	Contextual Relevancy	Measures how relevant the answer is to the specific context
Response	LLM-as-Evaluator	Another LLM model evaluates the quality of the response
	Answer Relevancy	Check whether the answer directly and accurately corresponds to the query
	Faithfulness	Check whether the answer is true in context and whether the LLM is not making up information
	Contextual Relevancy	Measures how relevant the answer is to the specific context

Experimental Results. RAG vs. MRAG

We evaluated both traditional RAG (Azure OpenAI Assistant) and mRAG using the GPT-4.1 model. The evaluation involved three main metrics: LLM as Evaluator, Answer Relevancy, calculated as the mean cosine similarity between the original question and a set of questions generated based on the answer; and Answer Semantic Similarity, measured as the cosine similarity between the expected and actual answers. The results of this evaluation are presented in Table 3.

Table 3—Experimental results for traditional RAG (Azure OpenAI Assistant) and mRAG.

Modality	Model	Contextual Precision	Contextual Recall	Contextual Relevancy	LLM as Evaluator	Answer Semantic Similarity	Answer Relevancy	Faithfulness
All	Traditional RAG (Azure OpenAI Assistant)	0.4568	0.9904	0.9904	0.3899	0.6274	0.7987	0.6137
	mRAG				0.9109	0.7298	0.7433	0.9273
Text	Traditional RAG (Azure OpenAI Assistant)	0.5027	0.9889	0.9889	0.4359	0.5862	0.7867	0.7180
	mRAG				0.9890	0.702	0.7400	0.9253
Table	Traditional RAG (Azure OpenAI Assistant)	0.4073	1.0000	1.0000	0.2500	0.6676	0.8381	0.4568
	mRAG				0.8722	0.7676	0.7710	0.9310
Image	Traditional RAG (Azure OpenAI Assistant)	0.4361	0.9737	0.9737	0.4333	0.7094	0.7170	0.6667
	mRAG				0.8758	0.8245	0.7307	0.9367

Overall, MRAG delivered the strongest performance, achieving the highest average scores in LLM as Evaluator (0.9109), Answer Relevancy (0.7298) and Faithfulness (0.9273). This indicates mRAG's stronger ability to produce contextually accurate and reliable answers.

The data clearly shows that mRAG approach has higher performance in Answer Semantic Similarity metric for all modalities. It means that actual output is close to the expected answer. Interestingly, the traditional RAG (Azure OpenAI) outperformed the mRAG in terms of Answer Relevancy. Since the Answer Relevancy metric was calculated based on cosine similarity between the original question and artificial questions generated from the answer, longer or more elaborated responses may have weaker direct lexical or semantic alignment, even if they are contextually appropriate.

When analyzing specific data modalities: text, tables, and images, the traditional RAG system displayed lower scores in evaluation metrics, including answer relevancy and faithfulness, suggesting challenges in handling diverse data types effectively.

In summary, mRAG approach demonstrates superior performance and robustness across various modalities, highlighting its effectiveness over the traditional RAG system for multimodal retrieval-augmented generation tasks.

Experimental Results. LLM Models Comparison

We benchmarked the multimodal RAG (mRAG) pipeline using five language models that differ in size and licensing:

- **GPT-4.1** (OpenAI, proprietary)
- **Llama-4-Maverik** (Meta, open-weight)
- **Mistral-Large** (Mistral A, commercial)
- **GPT-OSS-120B** (OpenAI, proprietary)
- **GPT5** (OpenAI, proprietary)

The evaluation results, where the models were assessed across multiple modalities: text, table, image, and video - are presented in [Table 4](#).

Table 4—Experimental results across different modalities comparing different models.

Modality	Model	Contextual Precision	Contextual Recall	Contextual Relevancy	LLM as Evaluator	Answer Relevancy	Faithfulness
All	GPT4.1	0.5148	0.9728	0.9708	0.9109	0.8691	0.9273
	LLAMA				0.9054	0.8563	0.9186
	MISTRAL				0.8651	0.8034	0.9254
	GPT-OSS-120B				0.9240	0.9704	0.8994
	GPT5				0.9341	0.9038	0.9459
Text	GPT4.1	0.5651	0.9579	0.9637	0.9890	0.8446	0.9253
	LLAMA				0.9692	0.8401	0.9416
	MISTRAL				0.9760	0.7307	0.9241
	GPT-OSS-120B				0.9987	0.9628	0.9243
	GPT5				0.9897	0.8888	0.9623
Table	GPT4.1	0.4821	0.9348	0.9468	0.8722	0.9141	0.9310
	LLAMA				0.8556	0.8151	0.9159
	MISTRAL				0.8370	0.8164	0.9144
	GPT-OSS-120B				0.8463	0.9458	0.8590
	GPT5				0.8667	0.8479	0.8638
Image	GPT4.1	0.4714	0.9705	0.9634	0.8758	0.8124	0.9367
	LLAMA				0.8939	0.8105	0.9444
	MISTRAL				0.7667	0.6559	0.9125
	GPT-OSS-120B				0.9394	0.9773	0.8941
	GPT5				0.9394	0.8570	0.9349
Video	GPT4.1	0.3199	0.9583	0.9583	0.7111	0.9192	0.9556
	LLAMA				0.7555	0.7351	0.9889
	MISTRAL				0.6333	0.8788	0.9175
	GPT-OSS-120B				0.8778	1.0000	0.9630
	GPT5				0.9333	0.9563	1.0000

GPT-5 achieved the highest average scores in LLM as Evaluator (0.9341) and Faithfulness (0.9459), confirming its advantage in producing contextually consistent and source-faithful responses. GPT-OSS-120B recorded the best performance in Answer Relevancy (0.9704), indicating strong capability in aligning output closely with the query. For image-based queries, GPT-5 and GPT-OSS-120B demonstrated superior integration of visual context compared to other models. All models performed well on textual inputs, with GPT-5 and LLAMA achieving the highest Faithfulness scores, and GPT-5 leading with 0.9623. Video remains the most challenging modality. The developed pipeline showed low Contextual Precision, highlighting the difficulty of accurately extracting key information from dynamic sequences. In the case of tabular inputs, the solution also achieved very high Contextual Recall scores (up to 1.000), confirming its ability to interpret and extract relevant insights from structured data.

In summary, mRAG shows the effectiveness in unifying embeddings across diverse modalities. GPT-4.1, GPT-5 and GPT-OSS-120B remain the most reliable models overall, GPT-OSS-120B and GPT-5 show competitive or superior performance in structured (table) and visual (image/video) tasks. Despite these gains, video remains the most challenging modality. Future work should focus on:

- Improving temporal embedding techniques for video keyframes.
- Research for more advanced techniques for video modality conversion into embeddings (that will include more dynamics to be caught).

Conclusions and Future Work

Agentic generative AI powered by multimodal retrieval-augmented generation is emerging as a transformative technology for the downstream and upstream oil and gas industry. By integrating heterogeneous unstructured multimodal sources - such as seismic surveys documentation, well-logs visuals, maps (geospatial and reservoir), and textual reports - these systems provide context-aware automation while aligning outputs with corporate knowledge bases and industry standards. Multimodal RAG accelerates oil and gas exploration by integrating seismic survey data, well logs, drilling records, and reservoir models with real-time knowledge retrieval to optimize prospect evaluation, reduce exploration risk, and improve subsurface characterization. In drilling and completions, it enables real-time rig status monitoring based on daily reports, helps with optimization of well placement, and improves planning through systematic use of lessons learned.

Beyond exploration and drilling, the applications of mRAG and Agentic GenAI extend across production, HSE, and sustainability domains. Production-focused agents integrate SCADA streams, allocation reports, and well tests to automate surveillance and generate actionable field development insights. Artificial lift surveillance agents, as an example, combine surface and subsurface signals with equipment performance histories to anticipate failures, adjust operating parameters, and optimize recovery. HSE-focused systems check that work permits are valid, compare safety plans with what's actually happening on site, and find the real causes of incidents; in combination with real-time CCTV footage processed by mRAG, they can improve automated compliance checks and speed up emergency response. Similarly, emissions agents integrate operational data, flaring logs, and abatement strategies to detect methane emissions, propose mitigation measures, and facilitate regulatory compliance.

Taken together, these advances illustrate how agentic, multimodal RAG systems are not merely incremental improvements but enablers of a paradigm shift toward data-driven, autonomous, and resilient oilfield operations. Their continued development and deployment promise significant gains in efficiency, safety, and environmental stewardship across the upstream value chain.

This work presents a comprehensive methodology for designing and implementing a Multimodal Retrieval- Augmented Generation (mRAG) system, with particular focus on pipelines for extracting, embedding, and indexing diverse data modalities, including text, tables, images, and videos. By integrating a unified vector space and leveraging the generative capabilities of modern LLMs, the proposed mRAG architecture enables more contextually relevant, accurate, and faithful responses across a wide range of input types.

Through extensive benchmarking across five foundation models: GPT-4.1, LLaMA-4-Maverik, Mistral-Large, GPT-OSS-120B and GPT-5 we demonstrated that mRAG delivers strong performance in handling complex, domain-specific multimodal queries. While text and tabular data are handled effectively by all models, video remains a key challenge, with low contextual precision observed across the board. Future work will focus on several key areas:

- Improving video understanding through advanced multimodal-temporal architectures and frame-level semantic alignment.

- Introducing mechanism of hybrid search and multi-level indexes to handle more efficiently information of the same nature (e.g. different drilling reports contain semantically very similar content. It may result in incorrect retrieval if only embeddings search is used).
- Using combined embeddings to encode multimodal inputs to the system (no separate processing pipelines for different modalities).

By addressing these directions, we aim to extend the capabilities of mRAG systems capable of understanding data-intensive domains such as oil and gas.

Nomenclature

D&C - Drilling and Completion
 GenAI - Generative Artificial Intelligence
 LLM - Large Language Model
 LWD - Logging While Drilling
 MRAG - multimodal RAG
 O&G - Oil and Gas
 RAG - Retrieval-Augmented Generation

References

- Aquino, G., Azevedo, N. et al 2025. From RAG to Multi-Agent Systems: A Survey of Modern Approaches in LLM Development.
- Cheng, Y., Zhang, C. et al 2024. Exploring Large Language Model based Intelligent Agents: Definitions, Methods, and Prospects. arXiv:2401.03428.
- DeepEval Open Source LLM Evaluation Framework, <https://deepeval.com/>, (accessed 28 July 2025).
- Drushchak, N., Polyakovska, N., Bautina, M., Semchenko, T., Koscielecki, J., Sykala, W., & Wgrzynowski, M. 2025. Multimodal Retrieval-Augmented Generation: Unified Information Processing Across Text, Image, Table, and Video Modalities. Proceedings of the 1st Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMaR 2025) (s. 59–64). Association for Computational Linguistics.
- Gao, Y. et al 2023. Retrieval-augmented generation for large language models: A survey. ArXiv, abs/2312.10997.
- Guu, K., Lee, K. et al 2020. Retrieval Augmented Language Model Pre-training. Proceedings of the 37th International Conference on Machine Learning, 13-18 July 2020, Virtual Event, volume 119 of Proceedings of Machine Learning Research.
- LangFuse Open Source LLM Engineering Platform, <https://langfuse.com/> (accessed 28 July 2025).
- Lewis, P. et al 2021. Retrieval augmented generation for knowledge-intensive NLP tasks," arXiv:2005.11401.v4. OpenCV documentation, <https://opencv.org/> (accessed 28 July 2025).
- Sun, L., Zhao, J. et al 2024. Fact-aware Multimodal Retrieval Augmentation for Accurate Medical Radiology Report Generation. ArXiv: abs/2407.15268.
- TinyDB 4.8.0 documentation, <https://tinydb.readthedocs.io/en/latest/>, (accessed 28 July 2025).
- Wu, K. et al 2024, Quantifying the tug-of-war between LLMs internal prior and external evidence. arXiv:2404.10198.
- Yin, S, et al 2023. A survey on multimodal large language models. ArXiv, abs/2306.13549. ArXiv, abs/2306.13549.
- Zhang, D. et al 2024. In Findings of the Association for Computational Linguistics ACL 2024, Bangkok, Thailand.